SOLUSI LATIHAN KARNAUGH MAPS

DDAK 2022/2023

By: Heri Kurniawan (Rev: 17-Feb-2025)

PERHATIAN
- Jangan baca solusi sebelum mengerjakan soal pada dokumen latihan
- Jika ada hal yang kurang jelas dari jawaban harap email ke: **herik@cs.ui.ac.id**

PART I

5-1. The K-map of a 3-variable function F is given below. What is the sum-of-minterms expression of F?

a \ bc	00	01	11	10
0	**1**	0	0	**1**
1	0	**1**	0	0

5-2. Draw the K-map for this function: $A(x,y,z) = x{\cdot}y + y{\cdot}z' + x'{\cdot}y'{\cdot}z$.

5-3. Identify the prime implicants and essential prime implicants for the K-maps below.

a.

a \ bc	00	01	11	10
0	**1**	**1**	0	**1**
1	0	**1**	0	0

b.

CD \ AB	00	01	11	10
00	**1**	**1**	0	**1**
01	0	0	**1**	**1**
11	**1**	**1**	0	**1**
10	**1**	0	**1**	**1**

5-4. Find the minimal SOP expression for $G(A,B,C,D)$, whose K-map is shown below.

CD \ AB	00	01	11	10
00	0	**1**	0	0
01	0	**1**	**1**	**1**
11	**1**	**1**	**1**	0
10	0	0	**1**	0

For questions 5-5, 5-6 and 5-7, refer to the following K-map of a function $W(A,B,C,D)$.

AB \ CD	00	01	11	10
00	**1**	**1**	0	**1**
01	0	**1**	**1**	0
11	0	**1**	**1**	**1**
10	0	**1**	0	0

5-5. Which of the following is not a prime implicant?

a. $A'\cdot B'\cdot C'$ b. $C'\cdot D$ c. $B\cdot D$ d. $A\cdot B\cdot C$ e. $A\cdot B\cdot D$

5-6 Which of the following is not an essential prime implicant?

a. $A'\cdot B'\cdot D'$ b. $C'\cdot D$ c. $B\cdot D$ d. $A\cdot B\cdot C$ e. $A'\cdot B'\cdot C'$

5-7 What is the minimal SOP expression for W' (the complement of W)?

a. $A\cdot B'\cdot C + A'\cdot B\cdot D' + A\cdot C'\cdot D' + B'\cdot C\cdot D$
b. $A\cdot B\cdot C' + A\cdot B'\cdot D' + A'\cdot C\cdot D' + B\cdot C'\cdot D$
c. $A\cdot B'\cdot C + A'\cdot B\cdot D' + A\cdot C'\cdot D + B'\cdot C\cdot D'$
d. $A\cdot B\cdot C + A\cdot B\cdot D' + A\cdot C'\cdot D' + B\cdot C'\cdot D$
e. $A\cdot B\cdot C + A'\cdot B\cdot D' + A\cdot C'\cdot D' + B\cdot C\cdot D'$

For questions 5-8, 5-9 and 5-10, refer to the following K-map of a function $Y(A,B,C,D)$.

AB \ CD	00	01	11	10
00	0	0	X	0
01	**1**	0	X	X
11	X	X	**1**	0
10	0	X	**1**	0

5-8. How many essential prime implicants (EPIs) are there?

a. 0 b. 1 c. 2 d. 3 e. 4

5-9. How many possible minimal SOP expressions for Y are there?

a. 0 b. 1 c. 2 d. 3 e. 4

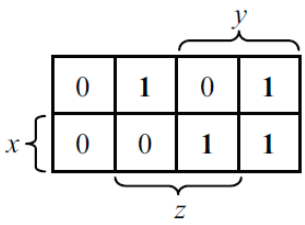
5-10. Which of the following is a minimal POS expression for Y?

a. $(A + B)\cdot(A + D')\cdot(A' + D)$
b. $(C + D')\cdot(B + D)\cdot(C' + D)$
c. $(A' + D)\cdot(B + D)\cdot(A + D')$
d. $(A' + D)\cdot(B + D)\cdot(C + D')$
e. All of the above.

JAWABAN PART I

5-1. $F(a,b,c) = a'\cdot b'\cdot c' + a'\cdot b\cdot c' + a\cdot b'\cdot c = \Sigma\, m(0,2,5)$.

5-2. $A(x,y,z) = x\cdot y + y\cdot z' + x'\cdot y'\cdot z$.



5-3. (a) 3 prime implicants: $a'\cdot b'$, $a'\cdot c'$, $b'\cdot c$.
2 essential prime implicants: $a'\cdot c'$, $b'\cdot c$.

(b) 7 prime implicants: $B'\cdot D'$, $A\cdot B'$, $B'\cdot C$, $A'\cdot C'\cdot D'$, $A\cdot C'\cdot D$, $A'\cdot C\cdot D$, $A\cdot C\cdot D'$.
4 essential prime implicants: $A'\cdot C'\cdot D'$, $A\cdot C'\cdot D$, $A'\cdot C\cdot D$, $A\cdot C\cdot D'$.

5-4. $G(A,B,C,D) = A'\cdot B\cdot C' + A\cdot C'\cdot D + A'\cdot C\cdot D + A\cdot B\cdot C$

5-5. (e) 5-6. (e) 5-7. (a) 5-8. (a)

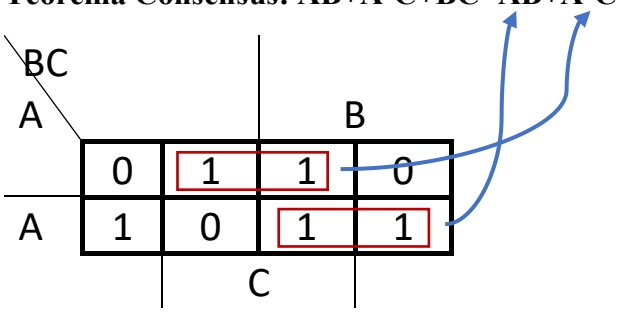
5.9. (e) [(1) $A'\cdot B\cdot D' + C\cdot D$; (2) $A'\cdot B\cdot D' + A\cdot D$; (3) $B\cdot C'\cdot D' + C\cdot D$; (4) $B\cdot C'\cdot D' + A\cdot D$]

5-10. (e)

PART II

5-12.* Draw an appropriate K-map to prove the Consensus Theorem.

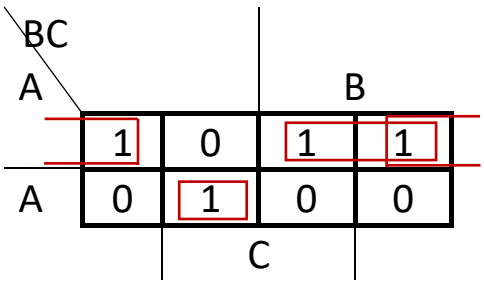
Teorema Consensus: AB+A'C+BC=AB+A'C



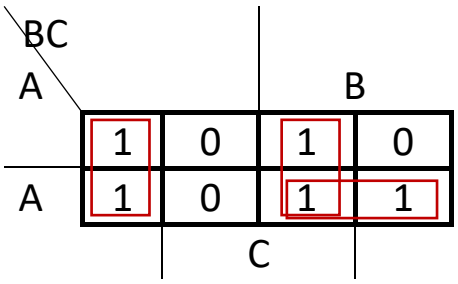
5-13.* Use K-map to find the minimal SOP expressions of the following functions.

(a) $f(A,B,C) = \Sigma\, m(0, 2, 3, 5)$
(b) $f(A,B,C) = \Sigma\, m(0, 3, 4, 6, 7)$
(c) $f(A,B,C,D) = \Sigma\, m(1, 2, 7, 12, 15) + d(5, 9, 10, 11, 13)$
(d) $f(A,B,C,D) = \Sigma\, m(0, 2, 5, 15) + d(8, 9, 12, 13)$

(a) F(A,B,C)=∑m(0,2,3,5)=A'C' + A'B + AB'C



(b) F(A,B,C)=∑m(0,3,4,6,7) = B'C' + BC + AB



(c) F(A,B,C,D)=∑m(1,2,7,12,15) + d(5,9,10,11,13) =BD + C'D + ABC' + B'CD'

AB \ CD				
	0	1	0	1
	0	x	1	0
A	1	x	1	0
	0	x	x	x

B, C, D

(d) F(A,B,C,D)=∑m(0,2,5,15) + d(8,9,12,13) =A'B'D' + BC'D + ABD

AB \ CD				
	1	0	0	1
	0	1	0	0
A	x	x	1	0
	x	x	0	0

B, C, D

5-14.* Find the minimal POS form for the following functions.

(a) $f(A,B,C,D) = \Pi\, M(4, 7, 9, 11, 12) \cdot D(0, 1, 2, 3)$
(b) $f(A,B,C,D) = \Pi\, M(0, 3, 7, 12) \cdot D(2, 10, 11, 14)$

(a) **F(A,B,C,D)=Π *M*(4,7,9,11,12). D(0,1,2,3)**

Pada KMAP, nilai 0 merepresentasikan maxterm dan 1 merepresentasikan minterm. Untuk mencari POS sederhana, gabungkan kotak-kotak bernilai 0 dengan 0 ataupun dengan x.

AB \ CD				
	x	X	x	X
	0	1	0	1
A	0	1	1	1
	1	0	0	1

B, C, D

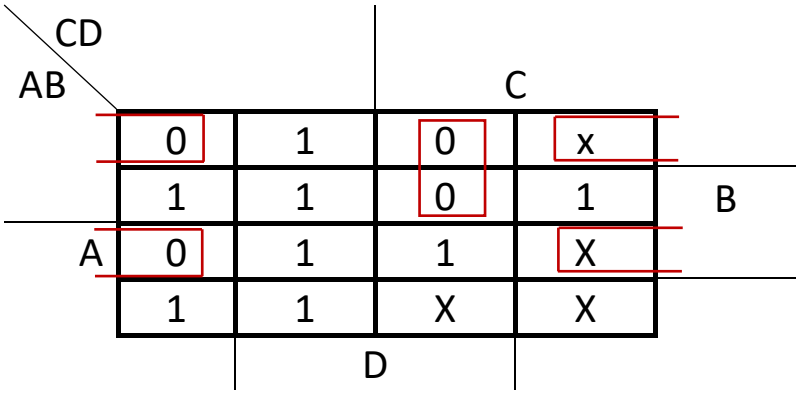
F'=B'D+A'CD+BC'D'

Konversi persamaan SOP ke POS dengan menggunakan Teorema DeMorgan

(F')'= (B'D+A'CD+BC'D')' = (B+D')(A+C'+D')(B'+C+D)

F(A,B,C,D)=(B+D')(A+C'+D')(B'+C+D)

(b) **F(A,B,C,D)=Π *M*(0,3,7,12). D(2,10,11,14)**



F'=A'B'D'+ABD'+A'CD ;

Konversi persamaan SOP ke POS dengan menggunakan Teorema DeMorgan

(F')'= (A'B'D'+ABD'+A'CD)' = (A+B+D)(A'+B'+D)(A+C'+D')

F(A,B,C,D)=(A+B+D)(A'+B'+D)(A+C'+D')

5-15. Design a BCD encrypting circuit that takes in a 4-bit BCD code *ABCD* and encrypts it into another BCD code *EFGH* using the following conversion table:

$0 \rightarrow 1$	$1 \rightarrow 6$	$2 \rightarrow 3$	$3 \rightarrow 7$	$4 \rightarrow 0$
$5 \rightarrow 8$	$6 \rightarrow 2$	$7 \rightarrow 9$	$8 \rightarrow 5$	$9 \rightarrow 4$

(a) Complete the truth table.
(b) Find the minimal SOP expressions for *E*, *F*, *G* and *H*.

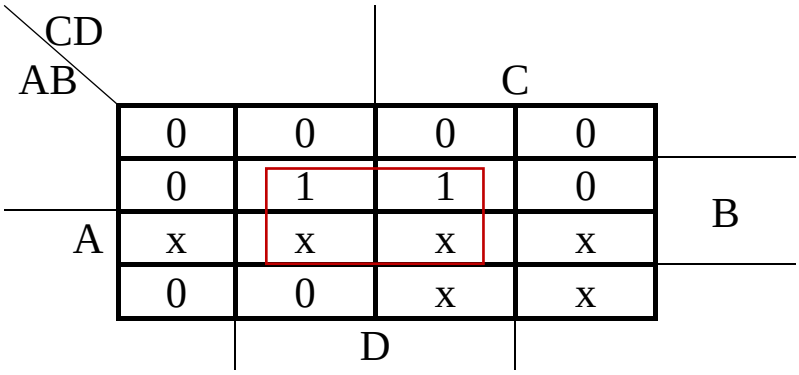
Materi tentang ini akan dipelajari lebih lanjut pada pembahasan Combinational Logic Design

(a) Truth table

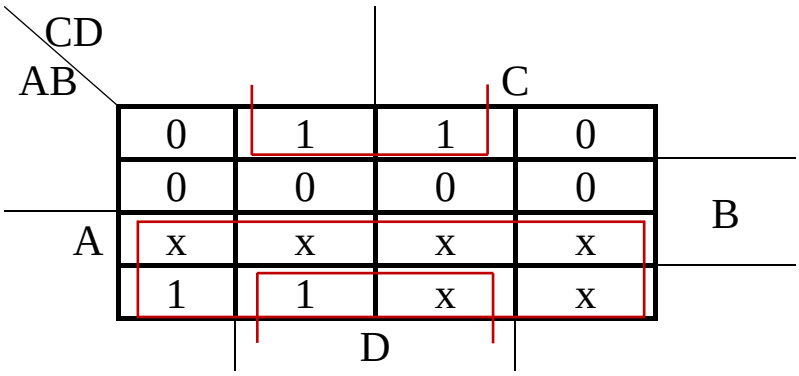
INPUT				OUTPUT			
A	**B**	**C**	**D**	**E**	**F**	**G**	**H**
0	0	0	0	0	0	0	1
0	0	0	1	0	1	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	1	1
0	1	0	0	0	0	0	0
0	1	0	1	1	0	0	0
0	1	1	0	0	0	1	0

0	1	1	1	1	0	0	1
1	0	0	0	0	1	0	1
1	0	0	1	0	1	0	0
1	0	1	0	X	x	x	x
1	0	1	1	X	x	x	x
1	1	0	0	X	x	x	x
1	1	0	1	X	x	x	x
1	1	1	0	X	x	x	x
1	1	1	1	X	x	x	x

E(A,B,C,D)=∑m(5,7)+d(10,11,12,13,14,15)
E(A,B,C,D)=BD



F(A,B,C,D)=∑m(1,3,8,9)+d(10,11,12,13,14,15)
F(A,B,C,D)= A + B'D



G(A,B,C,D)=∑m(1,2,3,6)+d(10,11,12,13,14,15)
G(A,B,C,D)= CD' + A'B'D

AB \ CD				
	0	1	1	1
	0	0	0	1
A	x	x	x	x
	0	0	x	x

B

C

D

H=∑m(0,2,3,7,8)+d(10,11,12,13,14,15)
H= B'D' + CD

AB \ CD				
	1	0	1	1
	0	0	1	0
A	x	x	x	x
	1	0	x	x

B

C

D

5-17.* Given the Boolean expression

$$F(W,X,Y,Z) = \Pi M(1,3,5,8,9,11,12) \cdot d(0,6,10,13)$$

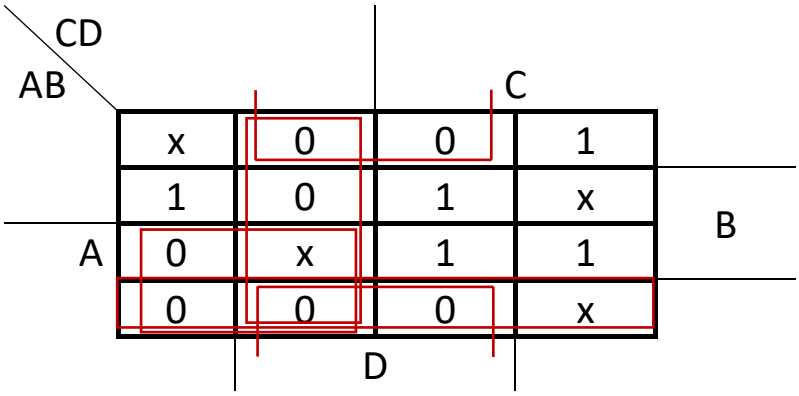
where 'd' represents don't-care.

(a) List out all the prime implicants in the K-map.

(b) List out all the essential prime implicants in the K-map.

(c) What is the simplified SOP expression for F?

(a) F(A,B,C,D)=Π *M*(1,3,5,8,9,11,12). d(0,6,10,13)



Prime Implicants:

B'D, AB', AC', C'D

Konversi Product Term ke Sum Term dengan menggunakan Teorema DeMorgan

(B'D)', (AB')', (AC')', (C'D)' → (B+D'), (A'+B), (A'+C), (C+D')

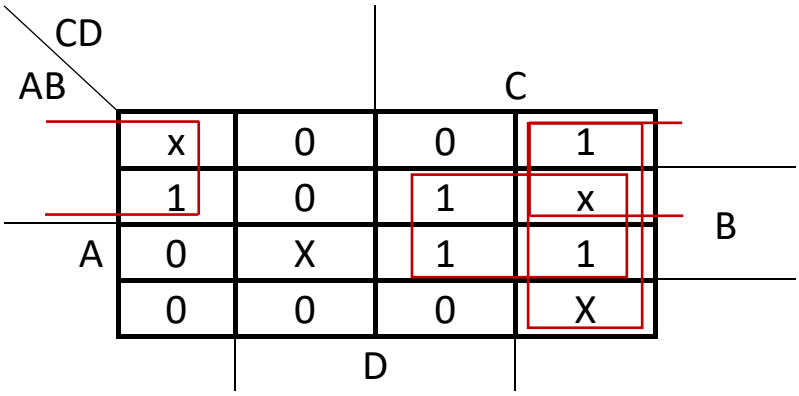
(b) Essential Prime Implicants:

(A'+C), (B+D'),(C+D')

(c) Fungsi POM' = SOM

F (A,B,C,D) = Π *M*(1,3,5,8,9,11,12). d(0,6,10,13)
F' (A,B,C,D) = Π *M*(2,4,7,15). d(0,6,10,13) sama dengan F(A,B,C,D) = ∑*m*(2,4,7,15) + d(0,6,10,13)

F(A,B,C,D) = ∑*m*(2,4,7,15) + d(0,6,10,13)



F(A,B,C,D)=A'D'+CD'+BC

5-18. Refer to the K-map for a Boolean function *K(A,B,C,D)* as shown below, where **X** represents don't care value.

X	0	0	**X**
X	**X**	**1**	**1**
0	**1**	**X**	0
X	**1**	0	**X**

(a) Express function *K* in big-*M* notation with don't care.

(b) How many prime implicants are there in the K-map? List them out.

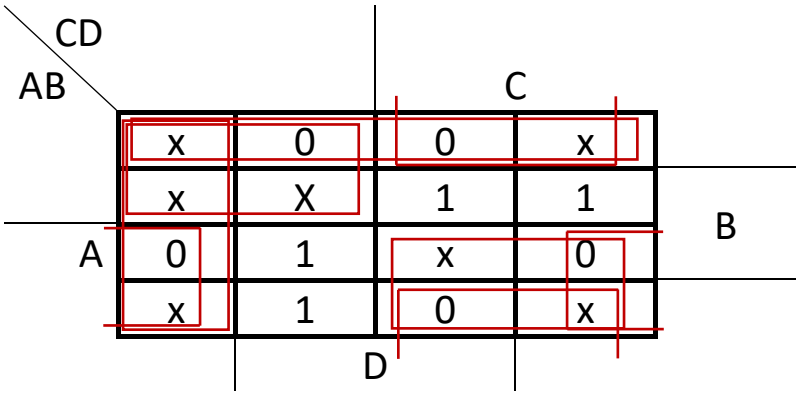
(c) How many essential prime implicants are there in the K-map? List them out.

(d) How many possible simplified SOP expressions for *K* are there? List them out.

(e) How many possible simplified POS expressions for *K* are there? List them out.

(a) F(A,B,C,D) = Π *M*(1,3,11,12,14). D(0,2,4,5,8,10,15)

(b) Prime Implicants (PI):



6 PI: C'D', A'C', A'B', AC, B'C, AD'

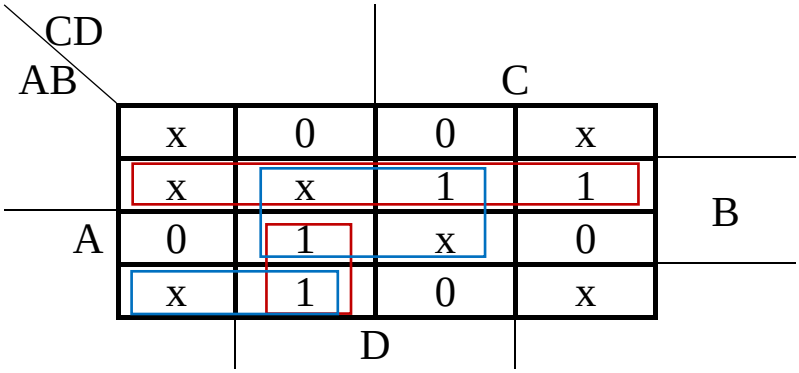
Terapkan DeMorgan law untuk mendapatkan versi POS:
(C+D), (A+C), (A+B), (A'+C'), (B+C')

(c) Essential Prime Implicants (EPI): 0 (tidak ada untuk fungsi POM)

(d) Beberapa ekspresi sederhana SOP yang dihasilkan berdasarkan K-MAP

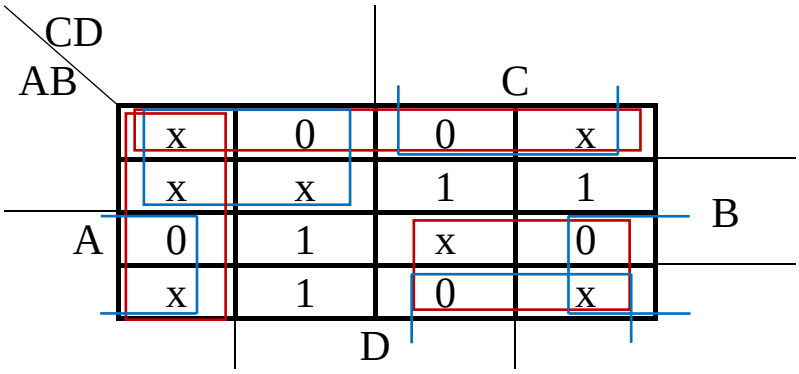
(Essential Prime Implicants: A'B' → ada untuk fungsi SOM)

1. **F(A,B,C,D)= A'B'+AC'D**
2. **F(A,B,C,D)= A'B'+BD+AB'C'**
3. **F(A,B,C,D)= A'B'+BD+ AC'D**



(e) Beberapa ekspresi sederhana POS yang dihasilkan berdasarkan K-MAP

Konversi persamaan SOP ke versi POS (ingat SOM'=POM)

1. **F'(A,B,C,D)=A'B'+C'D'+AC → F(A,B,C,D)= (A+B) (C+D)(A'+C')**
2. **F'(A,B,C,D)=A'C'+B'C+AD' → F(A,B,C,D)= (A+C) (B+C')(A'+D)**
3. **F'(A,B,C,D)=A'B'+AD'+AC → F(A,B,C,D)= (A+B) (A'+D)(A'+C')**

Tidak ada Essential Prime Implicants, sehingga persamaan hanya menggunakan Prime Implicants.